

Kanchan Bhattarai || Data Engineer

bhattaraikanchan07@gmail.com || 9202398866 || Milwaukee, WI, 53217, USA

LinkedIn: linkedin.com/in/Kanchan-bhattarai001

PROFESSIONAL SUMMARY

Data Engineer who specializes in developing and optimizing ETL pipelines, data warehousing, and real-time data processing using Python, SQL, Hadoop, and cloud platforms like AWS and Azure. I excel in collaborating with cross-functional teams to implement scalable data solutions, integrating machine learning models, ensuring data integrity, and providing technical mentorship. With a proven ability to transform complex business requirements into effective strategies, I am eager to leverage my expertise in big data technologies, cloud computing, and machine learning to drive innovation in data processing and analytics.

TECHNICAL SKILLS

Languages	: Python, SQL, Shell Scripting, Scala, Java, R, JavaScript, SnowSQL
Big Data Technologies	: Spark Hadoop, HDFS, Map Reduce, HBase, Apache Pig, Hive, Apache Impala, Yarn, Apache Flume, Kafka, Zookeeper.
Tools	: GIT , Github, Bitbucket, Docker, Kubernetes
Cloud Computing	: Amazon Web Services (lambda, S3, Glue, EC2, EMR, Redshift), Azure (Data Lake Gen2, Data factory, Azure SQL), Snowflake (Stage, Snow pipe, Snowpark), Databricks
Packages	: Numpy, Plotly, Seaborn, Matplotlib, Pandas, TensorFlow
Databases	: HBase, Cassandra, MongoDB, Oracle, SQL
Visualization	: Tableau, PowerBI, Excel

PROFESSIONAL EXPERIENCE

Data Engineer || Ford, Dearborn, MI

Mar 2023 – Present

- Efficiently migrated legacy systems to Azure Data Lake, resulting in a 15% reduction in operational costs. Executed seamless scalability to accommodate a 5x increase in users disaster recovery capabilities.
- Enhanced data quality and system integrity by creating Python utilities for thorough data validation between HDFS and source tables.
- Developed streamlined data import and export procedures among multiple databases including MySQL, SQL Server, Oracle, Teradata, and HDFS using Sqoop, resulting in an 85% improvement in database efficiency.
- Improved operational efficiency by automating complex ETL workflows, Oozie, integrating Sqoop and Hive actions.
- Developed efficient data pipelines using Apache Hive to optimize data warehousing, enabling seamless execution of complex queries and analytics.
- Applied knowledge of vehicle failures, failure chain reactions, and FMVSS safety standards for Ford vehicles to improve data accuracy and reliability in automotive data analytics projects.
- Developed and enforced data workflows using Azure Data Factory to streamline data movement and transformation processes across multiple sources and destinations.
- Enhanced performance and scalability of data lake storage solutions by implementing Azure Synapse Analytics, resulting in a 10% increase in data processing speed and efficiency.
- Implemented Python scripts to conduct thorough data audits and integrity checks, ensuring accuracy and consistency across various platforms.
- Partnered with diverse teams to develop and execute scalable data solutions, leading to improved data decision-making company wide.
- Implemented real-time data streaming solutions with Apache Kafka, enhancing data availability and system responsiveness.
- Integrated machine learning models into data pipelines, using PySpark, to enable predictive analytics and advanced data insights.
- Enhanced ETL processes with Apache NiFi, ensuring efficient data ingestion, transformation, and delivery, reducing processing times.
- Leveraged Azure DevOps for continuous integration and continuous deployment (CI/CD) of data pipelines, ensuring rapid and reliable deployment

Environment: Azure Data Lake, HDFS, MySQL, SQL Server, Oracle, Teradata, Sqoop, Oozie, Apache Hive, Azure Data Factory, Azure Synapse Analytics, Apache Kafka, PySpark, Apache NiFi, Azure DevOps

Data Engineer || Capital One, Chicago, IL**Jun 2021 – Feb 2023**

- Engineered serverless ETL pipelines using AWS Lambda, Python, and SQL, resulting in a 15% reduction in data processing costs, achieving a data accuracy rate of 95%, and efficiently managing 100GB of data daily.
- Enhanced data warehousing solutions with Amazon Redshift, optimizing query performance by 25% for real-time analytics for 500+ users, managing 500GB daily financial transactions.
- Created machine learning pipeline for predictive analytics, boosting model training speed by 35% and achieving 95% accurate forecasts on production data.
- Developed efficient data ingestion processes using AWS Kinesis, enabling real-time streaming, and processing of high-volume financial datasets.
- Enhanced efficiency of S3 storage system, leading to significant cost savings and enhanced data security
- Implemented AWS Step Functions to automate intricate workflows, enhancing operational efficiency and minimizing manual involvement in data processing tasks.
- Enhanced data security by implementing AWS IAM protocols, ensuring industry standard compliance and bolstering data integrity across all data processing operations.
- Implemented CI/CD pipelines using AWS CodePipeline to automate deployment procedures, facilitating swift and dependable updates for data applications.
- Conducted extensive data quality checks using AWS Glue DataBrew and Python scripts, enhancing data reliability and upholding high standards of data integrity.
- Collaborated with data scientists to deploy machine learning models in production using SageMaker, ensuring scalability and high performance of predictive analytics

Environment: Python, SQL, AWS Lambda, AWS Glue, Amazon Redshift, AWS Kinesis, S3, AWS Step Functions, AWS IAM, AWS CodePipeline, AWS CloudFormation, AWS Glue DataBrew, API Gateway, SageMaker

Data Engineer || T-Mobile, Columbus, OH**Aug 2019 – May 2021**

- Developed automation tools with Python, reducing manual tasks for 10+ Commodity Managers, improving business processes and operational efficiency significantly
- Implemented predictive maintenance systems in collaboration with cross-functional teams, leveraging machine learning algorithms to decrease downtime by 20% and lower maintenance costs by 10%.
- Automated comprehensive data quality checks with Python and SQL, reducing data anomalies by 15% and ensuring higher reliability and accuracy for business decision-making.
- Utilized large and varied datasets to perform quantitative research, developing tools that significantly improved process automation and business insights for commodity management.
- Engineered ETL processes using SQL and Python, streamlining data integration and transformation workflows, improving data processing efficiency across the organization.
- Optimized data storage and retrieval with SQL databases, ensuring efficient data management and reducing query times, enhancing overall system performance.
- Leveraged cloud platforms for scalable data solutions, ensuring cost-effective and flexible data storage and processing capabilities to support growing business needs.
- Implemented robust data security measures, ensuring data protection and compliance with industry standards, and safeguarding sensitive business information.
- Developed data visualization tools using Tableau, providing clear and actionable insights for stakeholders, facilitating informed decision-making and strategic planning

Environment: Python, SQL, Tableau, Apache Airflow, Apache Kafka, SQL databases, cloud platforms

EDUCATION**The University of Toledo || Toledo, OH****GPA: 3.88**

Bachelor's degree: Data Science, Computer Science